

DSAA3073: Theories in Data Science

Week 7 Open Investigation

Auditing a Boosted Fraud-Screening Rule

When does the AdaBoost training certificate survive a modified algorithm?

Basic analysis. A checkpoint for an insurance-claim screening system

Use only the AdaBoost update and empirical training-error results from the Week 7 Learning Sheet

An insurer uses binary decision rules to prioritize claims for manual fraud review. A development sample has been labeled after investigation. AdaBoost begins with uniform example weights and combines rules such as unusual claim timing, inconsistent documentation, and atypical provider history. The base learner is binary, labels lie in $\{-1, +1\}$, and every round is normalized exactly as in the Week 7 Learning Sheet.

During an audit, the team proposes the following provisional contract: at every round, the selected rule has weighted error at most $\varepsilon_t = 0.25$. The team wants a mathematical certificate that the empirical misclassification rate of the ensemble is at most one percent. It also asks whether satisfying that certificate is enough to deploy the model without prospective evaluation.

Basic Problem 1. Derive the round contribution and the required number of rounds

For realized errors $0 < \varepsilon_t \leq 0.25$, derive α_t and the normalization factor Z_t . Show first that the exact empirical certificate uses the realized product $\prod_t 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$. Then derive the worst-case product corollary implied by $\varepsilon_t \leq 0.25$ and the fixed-edge exponential corollary. At the boundary $\varepsilon_t = 0.25$, determine the next-round total mass assigned to examples that were correct and incorrect. Finally, find the smallest integer T that makes each worst-case corollary no larger than 0.01.

Basic analysis. Interpret the certificate without overclaiming

The theorem controls empirical classification error, not deployment risk

Basic Problem 2. Make the operational decision

Suppose the audit confirms the roundwise errors and the implementation for 37 rounds. State exactly what has been proved. Explain why neither the exponential-loss calculation nor zero empirical error establishes prospective fraud-detection performance. Recommend the next empirical step, taking account of the asymmetric costs of missed fraud and unnecessary manual review.

Thinking investigation. Formalize the certificate

The proof is a chain of an initial measure, a normalized update, exact round factors, and an edge condition

Express the analysis as

$$(D_1, \mathcal{U}, \mathcal{E}, \mathcal{C}),$$

where D_1 is the initial distribution on training indices, $\mathcal{U}$ the update and coefficient rule, $\mathcal{E}$ the edge sequence, and $\mathcal{C}$ the resulting empirical certificate. The Basic theorem uses uniform D_1 , the exact log-odds step, and a fixed positive lower bound on every edge.

Investigation 1. Replace uniform initialization by cost-sensitive weights

Let D_1 be any positive normalized distribution, chosen before training to emphasize costly fraud examples. Unroll the normalized update and derive the exact product identity. Determine the classification error quantity that is directly bounded. If the operational report still requires the ordinary unweighted training error, derive an analytic upper bound in terms of $d_{\min} = \min_i D_1(i)$.

Thinking investigation. Shrink every AdaBoost step

The product theorem survives, but the optimal round factor changes

Investigation 2. Introduce a learning-rate parameter

Keep the binary base hypothesis and weighted error ε_t , but use

$$\tilde{\alpha}_t = \nu \alpha_t, \quad 0 < \nu \leq 1,$$

where α_t is the usual log-odds minimizer. Determine whether the telescoping identity remains exact. Derive the new normalization factor $Z_{t(\nu)}$ as an analytic function of ε_t and ν . Compare it with $Z_{t(1)}$ and state which equality in the Week 7 theorem no longer holds when $\nu < 1$.

Thinking investigation. Allow the weak edge to decay

A fixed edge is sufficient for exponential decay, but it is not necessary for convergence of the varying-edge certificate

Investigation 3. Classify power-law edge sequences

Suppose the exact AdaBoost update is retained but the round edges satisfy

$$\gamma_t = ct^{-a}, \quad 0 < c < \frac{1}{2},$$

rather than a uniform positive lower bound. Use the varying-edge theorem to determine for which values of a its certificate tends to zero. Derive the leading form of the bound for $a < \frac{1}{2}$, $a = \frac{1}{2}$, and $a > \frac{1}{2}$. Explain precisely which Week 7 corollary fails and which theorem remains valid.

Synthesis. State the general Week 7 conclusion

Return to the fraud-screening system. For each modification, identify whether the original theorem is preserved, preserved with a changed target, or replaced by a new analytic expression. State one implementation quantity and one statistical quantity that the audit must record.